THE PRESIDENCY | FEDERAL REPUBLIC OF NIGERIA

SURE-P

SUBSIDY REINVESTMENT & EMPOWERMENT PROGRAMME

CCT BENEFICIARY DATA & PAYMENT MANAGEMENT SYSTEM

USER REQUIREMENTS LIST

FEBRUARY 2014

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EXECUTIVE SUMMARY

The CCT Programme requires data collection and beneficiary payment software, for use both in the central SURE-P MCH Project Implementation Unit (PIU) and by staff in the field. The software system will include sync with secure online (cloud-based) storage of the beneficiary data collected, requiring both offline and online modes for data entry.

The software and associated databases are to be used for the primary operational activities of:

- Entering CCT beneficiary data (personal data and records of services attended);
- Calculating and recording payments of the conditional cash transfer, according to the co-responsibilities each beneficiary fulfills;
- Producing summary figures and performing calculations for monitoring and evaluation purposes.

The application software, which should be tightly built, is required to therefore integrate seamlessly not only with the PIU's central database of SURE-P health facilities but also with the PIU's finance and accounts management system.

The objectives for developing this application are to:

- Automate and streamline the process of capturing and transmitting CCT beneficiaries' data in as close to real time as is operationally possible;
- Track compliance of pregnant women with CCT conditions;
- Establish a proxy system for identifying CCT beneficiaries;
- Track pregnant women's use of health commodities and services;
- Track rate of attrition within the CCT programme;
- Create a rich repository of beneficiary data that can be analyzed to identify status, trends, and key issues;
- Build a valuable source of beneficiaries' information that can be shared with external stakeholders for reporting and other purposes
- Enable automated calculation of the amounts beneficiaries are entitled to, based on the co-responsibilities they have fulfilled (see Table 1: Cash entitlement for CCT);
- Automate and streamline the process of generating payment schedules for the payment of cash support;
- Automate and streamline the process of recording the amounts disbursed to each beneficiary and the total amounts of cash support disbursed over time.

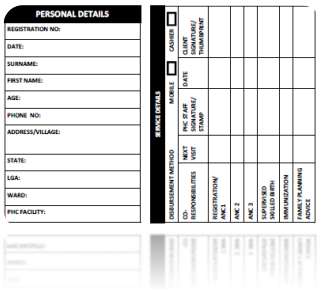
We aim to build a solution that will not only serve the current needs of the Project, but will also be robust enough to handle the CCT programme's scale up to 500 PHCs and beyond. This Requirements List documents what is expected from the application and sets out the associated minimum performance standards.

CCT DATA MANAGEMENT WORK FLOW



1. Pregnant woman arrives at PHC, registers her pregnancy and enrolls into the SURE-P CCT Programme.
2. Pregnant woman then comes to PHC at pre-scheduled intervals to receive health services





At every clinical encounter, health worker at facility updates the paper-based registration log/ consultation form with beneficiaries' service details.



CCT field staff move round facilities weekly, and (using **laptop or tablet**) enter manual data into electronic forms.





Data Clerk connects to the internet and transmits patient data to the PIU's cloud- based database.

Note that data entry is in most cases done offline, as not all field officers have mobile internet capacity, and many rural PHCs in any case have no network coverage.

GENERAL APPLICATION REQUIREMENTS

1.0 DATA ENTRY

1.1 **Online and offline modes for data entry.**

1.1.1 This is a *sine qua non* for the project as data entry will be conducted weekly by CCT field staff in PHC facilities, in areas where mobile data coverage is unreliable or absent. CCT field staff will be given modems to enable them to send data in the field, but due to poor coverage it is not possible for them to remain connected throughout their working day. A solely webform-based data entry solution will therefore lead to individual beneficiary records being lost after entry if the internet connection fails, or to field staff being unable to start work until they have a connection.

1.1.2 A non-browser-based application is required for data entry in the field (see above). The workflow envisaged is that CCT field staff enter data while in the field into this application, which will automatically 'push' saved data to the secure cloud-based database when connected to the internet.

1.1.3 This database should sync with all users' installed programmes whenever they connect to the internet.

2.0 INTEROPERABILITY

2.1 **Ability to import data from MS Excel.** Since CCT data collected in 2013 is held in spreadsheets and needs to be uploaded to the new database, this requirement is a *sine qua non*. In cases where data import will be required, fields on an Excel (.xls/.xlsx/.xlxm) worksheet should be mapped to corresponding fields in the application form.

2.2 **Ability to Export Data into Excel (.xls/.xlsx/.csv), Access (.mdb, .accdb):** This is required so that data within the application can be analyzed in ways not covered by the integrated report functions (see **5.2**, **9.0**, **10.0**, **11.0**).

2.3 **Ability to accept data from approved external applications**: This is required as some PHCs have tablets provided by development partners, which record beneficiary details (see 7.0) in clinical decision support applications. These will include all the data required for the CCT database and therefore it should be possible, with the technical input of partners providing these applications, for data recorded for CCT beneficiaries to 'push' to the cloud-based database as above (1.1.2).

3.0 DATA MANAGEMENT

3.1 **Search Capacity:** Ability to search beneficiaries by registration number, child's file number, name, location (facility; ward; LGA; and state), phone number, and address.

3.2 **Beneficiary Validation**

3.2.1 **(i):** The application should have the ability to query the entire database and flag possible duplicate registrations by women in each SURE-P cluster with any two or more of the following fields in common: name, age or DoB,

phone number, address. These will be reviewed by users with coordinator privileges (see **15.2**) to check if the shared details are coincidence or fraudulent duplicate registration.

3.2.2 **(ii):** The application should have the ability to query the entire database and flag for rejection any new entry indicating reenrollment by a beneficiary who exited the Programme within the last two years, *except* in cases of miscarriage, stillbirth or neonatal death within the last two years.

3.2.3 These flagged records should be stored in the cloud, but kept in a holding area where they are not visible to Viewer users (see **15.2**) and do not appear in reports and searches, until they are approved by an Updater, Coordinator or Administrator.

3.2.4 For Updater users (ie field officers) and state-level Coordinators using the web interface, the front page after login should include a list of potential false registrations in their area of responsibility (facility/state) flagged by the system.

3.3 **Beneficiary Tracking:** The application should be able to track active beneficiaries who missed a planned appointment date *or* who has not had attendance recorded for at least one month, who may have defaulted (see item **10.0**).

3.4 **Automatic Expiry Dates:** All beneficiary records should be set to automatically expire exactly nine months after the first entry. This means that a pregnant woman has to complete the programme within a maximum time of 9 months from the day she was first registered. Once this threshold is passed, the system should change her status to 'expired' and should initiate a pop-up that asks why the beneficiary did not complete the process before expiring.

4.0 INTEGRATION WITH CCT CASH DISBURSEMENT TO BENEFICIARIES

Table 1: Cash entitlement for CCT

Name of service	Corresponding CCT Co-Responsibility
Registration & ANC1	**1** **(=N= 1000)**
ANC2	**2** **(up to =N=1000, pro-rated)** =N= 300 for one attendance =N= 600 for two attendances
ANC3	
ANC4	
Skilled Birth Attendance	**3** **(=N= 2000)**
Post Natal Care	**4** **(=N=1000, not pro-rated)** EITHER PNC OR Immunization qualifies for this cash allowance.
First Immunization (BCG & OPV)	
Family Planning Advice	Not required for cash allowance, but promoted to beneficiaries to be used along with other post-natal services.

4.1 **Generation of Cash Transfer Schedule:** Although the CCT partner banks (Stanbic IBTC and Zenith Bank) may be users of the software system and receive Cash Transfer Schedules through this system, responsibility for generating and approving Schedules is held by the SURE-P MCH PIU and **not** the partner bank.

4.1.1 SURE-P MCH PIU generates a schedule of the amount due to each beneficiary to be paid, as in Table 1.

4.1.2 Partner banks are then required to transfer funds to cover each Cash Transfer Day from the SURE-P MCH account, and administer the disbursement of these funds based on the Schedule.

4.1.3 After each Cash Transfer Day, the relevant partner bank will provide the PIU with a report of the amounts disbursed to each CCT beneficiary listed in the Cash Transfer Schedule.

4.1.4 The software system should be able to interface with this report to update each beneficiary's record in one import.

4.2 **Individual Records of Cash Transfer Received:** Each beneficiary's personal record screen must show how much of her stipulated cash transfer has been disbursed to her, and the date of disbursement(s).

4.2.1 Each beneficiary is expected to receive either one or two tranches of cash support during her enrolment in the CCT, depending on how far throughout the continuum of care she meets the CCT co-responsibilities.

4.2.2 After she enrols and attends ANC1, she will be paid N1000 at the next available Cash Transfer day after her registration is logged and verified.

4.2.3 Beneficiaries who remain in the programme throughout the continuum of care will receive up to a further N4000 after delivery, once their adherence to other co-responsibilities has been logged and verified.

4.2.4 Any given Cash Transfer Schedule will therefore contain a mixture of recently registered beneficiaries and beneficiaries who have exited the programme. Any amount previously disbursed to individual beneficiaries must be offset against their total entitlement when drawing up the Cash Transfer Schedule, to avoid overpayment.

5.0 INTEGRATED MONITORING & EVALUATION FUNCTIONS

5.1 **Summary Key Performance Indicators:** The system should include a permanent and detailed dashboard of key performance figures, viewable a) in total and b) by state, facility and month (see item **9.0**).

5.2 **Output graphs and tables based on report queries**: The application should be capable of generating ad hoc reports and exportable lists based on searches by date, location, etc (see item **11.0**).

LIST OF DATA TABLES REQUIRED

6.0 DATA TABLES

6.1 There should be a 'Zone Table' listing the six zones

6.2 A State table listing states and their zones (zones will be picked up from **6.1**)

6.3 An LGA table listing LGAs, their states and their zones (States and Zones to be picked up from **6.1 and 6.2** respectively)

6.4 A Ward table listing wards, LGAs, States and Zones (latter details picked up from their respective tables)

6.5 A CCT beneficiaries' table that collects data from Form **7.0**.

6.6 A PHC table that lists all Primary Health Facilities under the SURE-P Programme.

6.7 A General Hospital Table that is linked to the PHC table through a one-to-many relationship[1].

CCT BENEFICIARY FORM

The following form will capture beneficiary's details and feed the resulting data into Table 6.5.

7.0 CCT Beneficiary Record: List of fields filled by CCT data entry staff

7.1 **PHC at which beneficiary is registered**[2]

7.2 **Unique CCT registration number**

7.3 **Date enrolled into the CCT Programme**

7.4 **Name**

7.5 **Phone Number**

7.6 **Address**

7.7 **Village (or Ward)**

7.8 **Next of Kin**:

- 7.8.1 **Name**
- 7.8.2 **Phone number**
- 7.8.3 **Address**
- 7.8.4 **Relationship with Beneficiary** (drop down list with options: sister; mother; other relative; husband; friend; neighbour; other)

7.9 **PHC services used:**

- 7.9.1 **ANC1**
- 7.9.2 **ANC2**
- 7.9.3 **ANC3**
- 7.9.4 **ANC4**
- 7.9.5 **Delivery with Skilled Birth Attendance**
- 7.9.6 **Post Natal Care**
- 7.9.7 **First Immunization (BCG & OPV)**
- 7.9.8 **Family Planning Advice**

[1] This linkage allows the application to replicate the SURE-P MCH Programme's cluster model

[2] The application will extract the relevant PHC details from **6.6.**

Some restriction of data entry should ensure dates are entered correctly. Each service from ANC1-Skilled Birth Attendance is in chronological order. It is possible that delivery may occur on the same day as ANC services are used. Fields for PNC, immunization, and family planning advice may be dated in any order, but in all cases the same or a succeeding day to Skilled Birth Attendance.

When beneficiary data is entered in the system, the next planned date of attendance is logged from the beneficiary's consultation card. Each service (see table below) has a column for recording planned date of attendance, and another for recording actual attendances. This two-column format for logging attendances is useful for tracking defaulters and observing how far beneficiaries are planning their attendances throughout the continuum of care.

Service	**Planned Visit**	**Service Used?**	**Date of Attendance**
ANC1	*Field disabled for ANC1*	☑	2014-01-20
ANC2	*2014-02-20*	☑	Autofills when box is checked with planned date, but can be changed by user
ANC3	*2013-03-20*	☐	
etc			

If the checkbox is not ticked manually but a date is entered in the right-hand column, the box should automatically be checked. If the user entering data checks the box but does not enter a date, the system should prompt them to enter a date when they save the record.

7.10 **Referral record**[3]**:** The fields should consist of:

- 7.10.1 **Check box** with title "Referred to Hospital?"
- 7.10.2 **Date of referral** (field becomes active if box checked)
- 7.10.3 **Stage of continuum of care** beneficiary was referred (field becomes active if box checked) – same list of fields as item 7.9.

7.11 **Child's PHC file number**

7.12 **Outcome:**

In addition to specifying the status, we require that the application documents the reason why a beneficiary left the Programme. This should be through a drop-down list of common reasons for easy summary of the most common reasons, with an 'Other' option allowing free text entry.

Drop down list with options:

- 7.12.1 **Unknown** [This is the default for any newly created beneficiary record, until otherwise changed]
- 7.12.2 **Completed Programme**
- 7.12.3 **Dropped out of programme**
- 7.12.4 **Deceased**

[3] When a beneficiary is referred, records of her continuing care are located at the General Hospital rather than the PHC she originally visited. CCT field staff must be aware at any moment of which beneficiaries are being seen at the hospital to accurately record compliance with co-responsibilities.

7.12.5 **Miscarriage**
7.12.6 **Live birth**
7.12.7 **Still birth or neonatal death**

8.0 CCT Beneficiary Record: Fields dependent on other data entry

8.1 **Status (Active, Has Completed Care, Has Exited the Programme)**
Status is linked to 'position within continuum of care' (item 7.9) and 'outcome' (item 7.12) as recorded by field officers:

8.1.1 If a beneficiary received any postnatal care (7.9.6-7.9.8), or has options 7.12.2-7.12.7 selected above, and has a zero outstanding balance (item 8.4), her status should automatically change to 'Exited the Programme'
8.1.2 If she has received postnatal care, or has options 7.12.2-7.12.7 selected above, but has outstanding balance, it is 'Has Completed Care'.

8.2 **Total Cash Entitlement**
Calculated pro rata according to **Table 1.**

8.3 **Cash received**
Imported from payment reports uploaded after cash disbursement (see item **4.0).**

8.3.1 Cash received (i)
8.3.2 Date of receipt
8.3.3 Cash received (ii)
8.3.4 Date of receipt

8.4 **Total balance outstanding**

REPORTS

The application should be capable of generating both routine and ad hoc (customizable) reports. Some types of reports we will require are:

9.0 Routine reports and summary figures (Performance Dashboard)

9.1 The online database should be able to sum the services attended and produce overall figures of:

9.1.1 **Number attending** each service in the continuum of care (item **7.9**)
9.1.2 **Rate of increase** in attendance from baseline[4]
9.1.3 **Percentage of beneficiaries who attended** for the full continuum of care and are thus entitled to N5000
9.1.4 **Percentage of beneficiaries** defaulting from programme[5]

[4] Calculated from static baseline monthly average attendance figures per facility; to be provided.
[5] Defined as reaching 9 months after enrollment (see item 3.4) without having completed the continuum of care, *ie* without having earned the N1000 for co-responsibility 4.

9.1.5 **Percentage of women retained** between any two stages of the continuum of care (item **7.9**) (e.g. % of those who delivered with skilled attendance, who then stayed in the programme for post-natal care and/or immunisations).

9.1.6 **Percentage of women who have received** all or part of their cash entitlement

9.1.7 **Percentage of beneficiaries who have exited** the programme after receiving their full cash allowance

9.2 Each one of summary figures should be available to generate tables and graphs restricting the figures by:

9.2.1 **Facility**

9.2.2 **LGA**

9.2.3 **State**

9.2.4 **Zone**

9.2.5 **Month**

9.2.6 **Year**

NB: Examples of the offline dashboards currently used will be provided to give definitions of the figures to be extracted from the database.
A good example of an existing dashboard using live project data is located at the NPHCDA Performance Based Financing online portal: https://nphcda.thenewtechs.com/data.html

9.3 The dashboard will update automatically based on standard queries to the database, so it provides an up-to-date summary of CCT activity at the time it is viewed.

9.4 The dashboard (**9.1.1-9.1.7**) should be available to all viewers, including who do not have the security access to view individual beneficiary records (see item **15.2**). For Viewers, Administrators and national-level Coordinators, the front page after login should include a table of the summary figures listed in **9.1.1,** both **national totals** and for the **state(s)** they are responsible for.

10.0 Routine beneficiary tracking

10.1 The application should enable tracking of potential defaulters, ie active beneficiaries who:

10.1.1 Missed a planned appointment date *or*

10.1.2 Have not had attendance recorded for at least 30 days

These beneficiary records require attention by staff at the state level, to determine if the women have permanently exited the programme (e.g. due to miscarriage or death), and to remind them to attend the PHC if they are still eligible to attend CCT services.

10.2 The database should incorporate a routine report of potential defaulters, which can be filtered according to the following fields:

10.2.1 State

10.2.2 LGA

10.2.3 PHC
10.2.4 Beneficiary address
10.2.5 Date of registration
10.2.6 Most recent service attended
10.2.7 Date of most recent service attended

10.3 This Tracking function should be available to all user levels except for Viewers (see item **15.2**). For Updater users (ie field officers) and state-level Coordinators using the web interface, the front page after login should include a list of potential defaulters flagged by the system.

11.0 Ad hoc reports from customizable queries:

11.1 **Retrieval of Beneficiaries' Details**. Ability to extract details of CCT beneficiaries and view in a customizable summary table, or export to MS Excel/Access.
11.1.1 The user will check the fields she or he wishes to include in the report.
11.1.2 The options for fields to view will have to include all fields in the individual beneficiary record: name, phone number, address, payment due, date of birth or age, list of services attended and corresponding dates, services with missed appointment, etc

11.2 **Location Search:** Ability to retrieve details of beneficiaries in a particular:
11.2.1 PHC facility
11.2.2 Cluster
11.2.3 Ward
11.2.4 LGA
11.2.5 State
11.2.6 Zone

We should be able to retrieve, for instance, the details of all CCT beneficiaries in:
11.2.7 Zango Kataf Local Government
11.2.8 Cluster 2 in Ogun State
11.2.9 Bayelsa State
11.2.10 South East zone

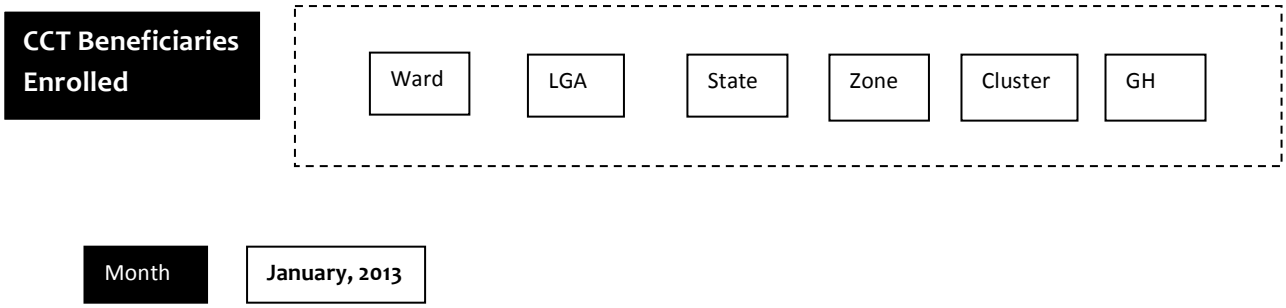
11.3 **Status of Beneficiary**: Application should capture beneficiary's status (see item 7.12) and be able to search by this status.

11.4 Number (Count) of beneficiaries in a particular geographical location or time frame
11.5 Details of beneficiaries nationwide
11.6 Details of all beneficiaries at a particular point within the continuum of care[6]
11.7 Details from **11.6** mixed with restrictions of location and time[7]
11.8 Details from **11.2 and 11.3** mixed with restrictions of time frame
11.8.1 Month

[6] This report will draw data from Item **7.9.6**.
[7] We may want to know, for example, who are the CCT Beneficiaries who enrolled through Ifo PHC in the month of February 2013?

11.8.2 Quarter
11.8.3 Year
11.8.4 Since a specific date in the past[8]
11.8.5 Between two specific dates in the past.

SAMPLE REPORT FORMAT[9]



USER INTERFACE

12.0 USER INTERFACE

12.1 We require a well-designed, clean and uncluttered user interface that is easy to navigate. In terms of navigation, we require that **all menus be accessible with three clicks or less.**

12.2 **Ability to set default location** for updaters (see **15.2**)

12.2.1 Data is entered in batches by CCT staff who visit facilities periodically. In a single login session all records entered can therefore be expected to be located in a single facility.

12.2.2 It should be possible to set a particular PHC as the location of all new records for the duration of a login session, or until that location is cleared by the user.

12.3 The developer will **create user documentation** (install instructions; user guide; FAQ) which will be available to all users of the programme and also hosted on the web server.

[8] We may want to answer a question like 'how many beneficiaries have we paid in Karu LGA since July 2012?
[9] Once we click the name of a state for instance, we want to see the details of all CCT beneficiaries in that state, arranged by PHC and Cluster.

INTEGRATION WITH THE PIU'S EXISTING SYSTEMS

13.0 INTEGRATION WITH EXISTING SYSTEMS

13.1 The application should integrate seamlessly with the PIU's central database such that all details of CCT beneficiaries are readily accessible through one central location.

13.2 In addition to this, the application should also provide for an interface through which it can communicate with the PIU's finance and accounts management system.

13.3 The technical details of these integrations will be worked out with the developer.

BACK UP & SECURITY

14.0 BACK-UP:

14.1 The application should have a built-in back up mechanism that backups data into a secure, cloud-based location.

15.0 SECURITY:

15.1 We require a multi-layer security architecture for the software application.

15.2 This should be implemented in such a way that every user is assigned an authorization/access level that is dependent on the role he/she is expected to play. This requires that each user account be associated with a default access level (see Table 2: Categories of access level).

Table 2: Categories of access level

GLOBAL ACCESS LEVEL	CAPABILITIES ASSOCIATED WITH LEVEL	USERS TO BE ASSIGNED TO ACCESS LEVEL
VIEWER	Viewers will have the capacity to view general information about CCT beneficiaries. They will be unable to add, edit or delete beneficiaries' records. Their ability to run queries and generate reports will also be limited.	Users to be assigned to this access level will include: 1. External stakeholders that need to be informed about the progress of the CCT programme. 2. Staff within the PIU with responsibility for monitoring the performance for the CCT programme.
UPDATER	Updaters will be able to add, edit and update records within the application. Two restrictions will, however, apply: 1. Updaters will be able to view and access only the records within their own state. 2. An Updater will not have the capacity to delete a beneficiary's record.	Users to be assigned to this level include CCT Field Officers at the facilities.
COORDINATOR	A coordinator will be a Viewer with special privileges. He/She will be able to view all records generated within his/her area of jurisdiction. In addition to this, a Coordinator should also have the capacity to flag a record as incorrect or inconsistent.	Users in this category will include: 1. Staff with responsibility for managing the CCT Programme at the national level 2. CCT State Technical Officers & CCT Field Supervisors 3. SURE-P MCH State Focal Persons
ADMINISTRATOR	An Administrator can view, edit and manage beneficiaries' information across all PHCs. He or she should also be able to create other categories of users and assign access levels to them as required.	Users in this category will include: 1. SURE-P MCH CCT Data Manager 2. SURE-P MCH IT Head 3. Other users as may be required.